

Problem

The measured open-circuit voltage across a certain amplifier is 16 V. The voltage drops to 8 V when a 8-Ω loudspeaker is connected to the amplifier. Calculate the voltage when a 24-Ω loudspeaker is used instead.

Step-by-step solution

Step 1 of 2

Consider the circuit diagram.

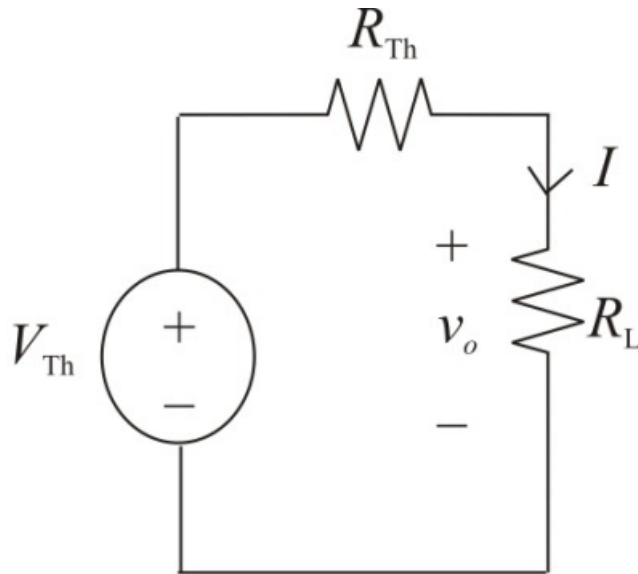


Figure 1

Step 2 of 2

When $V_{Th} = 16 \text{ V}$, the value of the voltage v_o is equal to 8 V which is half of the Thevenin's voltage. This occurs only when the load resistance is equal to the Thevenin's resistance.

Therefore, the value of the Thevenin's resistance is, 8 Ω.

Calculate the value of the voltage across a 24 Ω load.

$$v_o = (V_{Th}) \left(\frac{R_L}{R_L + R_{Th}} \right)$$

Substitute 24 Ω for R_L , 8 Ω for R_{Th} and 16 V for V_{Th} in the equation.

$$\begin{aligned} v_o &= (16) \left(\frac{24}{24 + 8} \right) \\ &= (16) \left(\frac{24}{32} \right) \\ &= \frac{24}{2} \\ &= 12 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_o for 24 Ω load is 12 V.